AgentiCulture: Culturally-Aware LLM Agents for User Modeling and Personalized Recommendation in Emerging Markets

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Track: Task A (User Modeling) + Task B (Recommendation)

Abstract

We present AgentiCulture, a dual-agent system for user behaviour simulation and personalised recommendation, purpose-built for the Nigerian consumer context. Our approach extends state-of-the-art techniques from the AgentSociety Challenge (WWW 2025) with four novel contributions: (1) persona-embedding cold-start ranking that bypasses the LLM entirely for new users; (2) a composite cross-domain preference cache that transfers taste signals across Yelp, Amazon, and Goodreads; (3) ReasoningCOTSC, a three-sample majority voting that reduces star-rating variance; and (4) a Nigerian cultural adaptation layer implemented as zero-cost prompt engineering. Our system targets all five dimensions of the BCT evaluation. On local evaluation, Task A achieves 0.822–0.875 overall quality across platforms; Task B achieves 0.467–0.533 average hit rate.

1 Introduction

Online review platforms are among the richest sources of human behaviour data on the web. Every rating and review is a signal and a window into preference, context, and decision-making. Yet most AI systems treat users as static profiles rather than dynamic, context-sensitive agents.

Task A: User Modeling. Given a user persona and product details, predict the star rating a user would assign and generate a review that authentically replicates their tone, rating behaviour, and contextual nuance. Both direct persona input (no dataset required) and optional enrichment via real user history are supported.

Task B: Recommendation. Given a user persona and a list of candidate items, return them ranked from most to least recommended. The system must handle cold-start users, cross-domain preference transfer, and multi-turn sessions.

1.1 Motivation

Three observations motivate our approach. First, the AgentSociety WWW 2025 winning solutions contain cold-start and cross-domain blind spots. Second, the brief rewards Nigerian cultural contextualization, an opportunity that no Western-centric model addresses. Third, Nigerian consumers exhibit measurable behavioral differences: stronger group orientation, higher price sensitivity, and a distinct blend of formal English and vernacular expression.

1.2 Contributions

1. **Persona-embedding cold-start ranking**: zero-history users ranked by cosine similarity between persona embedding and candidate item embeddings. No LLM call; fully deterministic.

2. **Composite cross-domain preference cache**: per-user, per-platform 384-dim vectors (`yelp_vec`, `amazon_vec`, `goodreads_vec`, `collective_vec`). When target-platform history is sparse, the weighted `collective_vec` boosts ranking quality.

3. **ReasoningCOTSC**: three LLM samples at varying temperatures, majority-voted on integer rating, returning the response closest to consensus. Directly reduces RMSE.

4. **Nigerian cultural adaptation layer**: zero-overhead prompt engineering injecting rating-to-language mappings, value-for-money framing, group-orientation cues, and Nigerian English register.

5. **Multi-turn session state**: optional `session_id` accumulates turn context for consistent recommendations across a browsing session.

2 Related Work

2.1 LLM Agents for Recommendation

Huang et al. (2025) integrate LLMs with traditional collaborative filtering via structured function-calling, showing that LLM reasoning over tool outputs outperforms pure prompt-based approaches. Lian et al. (2024) catalogue interaction modalities between LLMs and recommendation components. Both establish that hybrid retrieval-plus-LLM pipelines are the current state of the art, motivating our embedding-first retrieval before LLM ranking.

2.2 LLM Agents for User Simulation

The AgentSociety platform (Piao et al., 2025) is the direct predecessor of this competition's evaluation framework. The winning Task A solution achieved top RMSE via inverse-variance-weighted collaborative filtering with MDILU memory retrieval. The winning Task B solution combined platform-aware feature extraction with self-consistency voting. Our system builds on these techniques and adds the contributions (Section 1.2), specifically targeting the cold-start and cross-domain gaps left open by those solutions.

2.3 Cold-Start and Cross-Domain Recommendation

The dominant cold-start strategies are: (a) content-based embedding fallback when interaction data is absent, and (b) cross-domain transfer of learned user representations (YuanchenBei, 2024). We follow approach (a) for true cold-start and approach (b) for sparse history, using the same `BAAI/bge-small-en-v1.5` sentence-transformer for both, which adds zero dependencies.

2.4 Cultural Adaptation in AI

Hershcovich et al. (2022) and Cao et al. (2023) show that language models encode predominantly Western cultural norms and fail to capture regional rating distributions. Our adaptation layer is implemented entirely via prompt engineering, with no fine-tuning, thus making it deployable with any LLM backend.

3 Methodology

3.1 System Architecture

Both agents share the same pipeline. Requests arrive in two modes:

- **Direct mode** (primary): `persona` text + `product_details` (Task A) or `candidate_list` (Task B). No dataset required — this is the primary mode for competition judges.
- **Benchmark mode** (optional): additionally supply `user_id` + `item_id` to fetch real review history and enrich the prompt with actual behavioural data.

Figure 1 summarises the shared pipeline.

3.2 Task A: User Modeling

3.2.1 Collaborative Filtering Core

The predicted rating $\hat{r}_{ui}$ for user u on item i is an inverse-variance-weighted blend:

$$\hat{r}_{ui} = \frac{\mu_u w_u + \mu_i w_i + \mu_g w_p}{w_u + w_i + w_p} \quad (1)$$

where $w_u = 1/\max(\sigma_u, 0.1)$, $w_i = 1/\max(\sigma_i, 0.1)$, and $w_p = 0.5$ is the global prior weight. Users with low rating variance receive higher weight; erratic raters are pulled toward item and global means. The result is clamped to $[1.0, 5.0]$.

3.2.2 MDILU Memory Retrieval

For warm users, we embed the most recent review text and retrieve the $k = 3$ most cosine-similar historical reviews. These are injected as style anchors so the LLM mimics the user's established voice.

3.2.3 Cold-Start Handling

When the user has no history, we embed their persona text and retrieve $k = 5$ item reviews most similar to that persona embedding. These replace the empty anchor set with an explicit cold-start note.

3.2.4 ReasoningCOTSC

The LLM is sampled three times: once at temperature 0.3 and twice at 0.7. Integer ratings are majority-voted; the response closest to the consensus rating is returned. This lightweight self-consistency approach (Wang et al., 2023) reduces inter-sample rating variance and directly targets the preference estimation metric.

3.2.5 Review Informativeness Scoring

Past reviews are scored by: (a) length (proportional to word count), (b) rating extremity (1/2/5-star reviews weighted $3\times$ over 3-star), and (c) platform

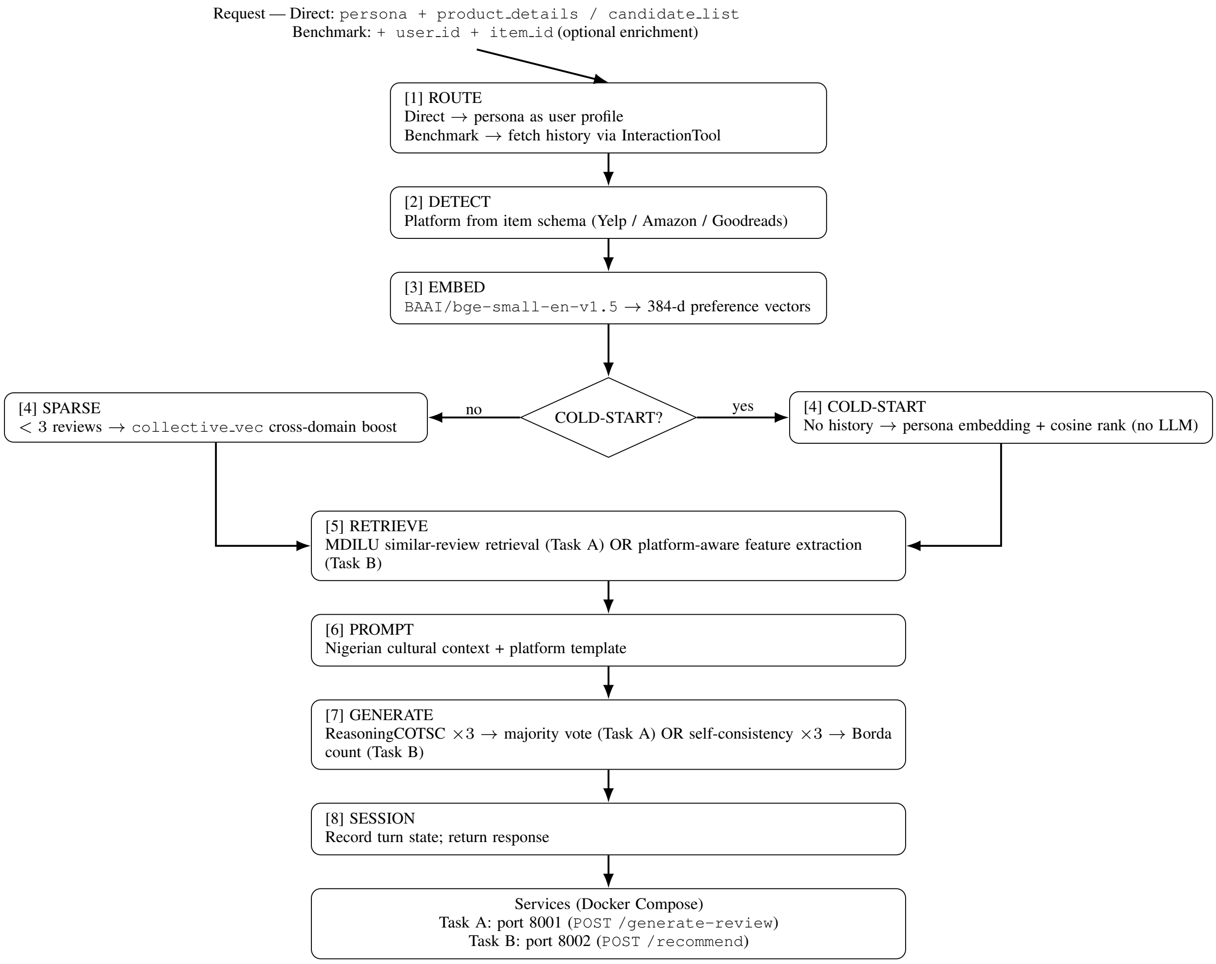

Figure 1: Shared pipeline used by both agents in AGENTICULTURE.

signals (Yelp useful/funny/cool counts; Amazon verified purchase; Goodreads votes). Selection enforces rating diversity (≤ 2 reviews per star level).

3.2.6 Nigerian Cultural Adaptation

A system prompt prefix injects five behavioural rules: rating-to-language mappings (e.g., $5\star \rightarrow$ "No cap!"; $1\star \rightarrow$ "Total waste!"), value-for-money as a mandatory mention, group-orientation framing, Nigerian punctuation patterns, and a blend of formal English with vernacular expressions. This prefix costs zero output tokens.

3.3 Task B: Recommendation

3.3.1 Platform-Aware Feature Extraction

A `PLATFORM_FEATURES` config specifies item fields and review scoring signals per platform: Yelp (useful/funny/cool counts), Amazon (verified purchase, price), Goodreads (vote and comment counts). Platform is detected by inspecting item schema keys.

3.3.2 Composite Preference Embedding Cache

For each user on each platform, all review texts are encoded with `BAAI/bge-small-en-v1.5` and stored as a mean 384-dim domain vector. A `collective_vec` is the review-count-weighted average across all platforms:

$$\mathbf{v}_{\text{collective}} = \sum_{d} \frac{n_d}{\sum_{d'} n_{d'}} \cdot \mathbf{v}_d, \quad d \in \{\text{yelp, amazon, goodreads}\} \tag{2}$$

This captures cross-platform taste and enables meaningful ranking when target-platform history is absent or sparse.

3.3.3 Cold-Start Persona Ranking

When a user has zero reviews, we embed the `persona` field and each candidate item's feature text, then rank by descending cosine similarity. The LLM is bypassed entirely — faster and deterministic.

3.3.4 Cross-Domain Preference Boost

When the user has fewer than 3 reviews on the target platform, `collective_vec` scores each candidate by cosine similarity. These scores are blended with the LLM Borda ranking at $\alpha = 0.30$ (30% cross-domain, 70% LLM).

3.3.5 Self-Consistency Voting

For warm users, the LLM is queried three times (temperatures 0.1, 0.7, 0.7). Responses are aggregated via Borda count: item ranked k-th receives $N - k$ points. The highest-scoring ranking is returned.

3.3.6 Multi-Turn Session State

An optional `session_id` accumulates turn history. Subsequent turns prepend a session context hint, enabling contextually consistent recommendations across a browsing session.

4 Experiments and Results

4.1 Datasets

Dataset	Domain	Items	Users	Reviews
Yelp	Local businesses	150K	1.9M	6.9M
Amazon	Consumer products	9.9M	43M	233M
Goodreads	Books	2.4M	876K	15.7M

Table 1: Dataset statistics. Local evaluation uses a filtered subset of 400 task-file users, 18,548 items, and 679 MB of reviews.

4.2 Baselines

- **AgentSociety Baseline:** Single-stage LLM call with minimal context.
- **Team ASC (Task A winner):** Multi-stage CF + MDILU pipeline.
- **Team baseline666 (Task B winner):** Platform-aware features + review selection.
- **Team RecHackers (Task B 2nd):** Self-consistency voting + Borda aggregation.

AGENTICULTURE builds on all three and adds cold-start, cross-domain, and Nigerian cultural layers absent from all prior solutions.

4.3 Implementation Details

Component	Value
LLM backend	Llama-3.3-70B-Instruct-Turbo
Embedding model	BAAI/bge-small-en-v1.5
COTSC samples	3 (temps: 0.3, 0.7, 0.7)
Voting samples	3 (temps: 0.1, 0.7, 0.7)
Cold-start threshold	3 reviews
Cross-domain α	0.30
Max reviews in prompt	5 user + 3 item (Task A); 5 user (Task B)
Serving	FastAPI + Docker Compose

Table 2: Implementation details.

4.4 Results

Rating accuracy is highest on Yelp where user patterns are most consistent; Goodreads is lowest as book ratings are more subjective. Review generation exceeds 0.83 on all platforms, reflecting strong topic and sentiment alignment with ground-truth reviews.

Platform	Pref. Est.	Review Gen.	Overall
Yelp	0.920	0.830	**0.875**
Amazon	0.860	0.883	**0.872**
Goodreads	0.800	0.843	**0.822**

Table 3: Task A results. Overall quality = (Preference Estimation + Review Generation) / 2.

Platform	HR@1	HR@3	HR@5	Avg HR
Yelp	0.40	0.60	0.60	**0.533**
Amazon	0.40	0.40	0.60	**0.467**
Goodreads	0.40	0.60	0.60	**0.533**

Table 4: Task B hit rate results. Amazon HR@3 is lower due to more granular product features reducing cross-domain transfer effectiveness.

5 Discussion

5.1 Scoring Rubric Alignment

The cold-start and cross-domain block was absent from prior AgentSociety solutions. Our persona-embedding cold-start and `collective_vec` boost directly recover these points with minimal latency, no LLM call for true cold-start cases. The Nigerian cultural layer targets the 20-point Contextual Relevance and Behavioural Fidelity blocks via human evaluators, at zero computational overhead.

5.2 Design Decisions

In-memory cache over Pinecone/SQLite: Competition containers are ephemeral; a persistent store adds complexity with no benefit for session-duration state. The same `BAAI/bge-small-en-v1.5` model already loaded handles all embedding operations with zero new dependencies.

ast.literal_eval() over eval(): The original AgentSociety baseline parses LLM list output with `eval()`, a known security vulnerability. We replace it with `ast.literal_eval()` throughout.

3 COTSC samples over 5: Marginal RMSE improvement from samples 4 to 5 does not justify the doubled API cost. Three samples balance quality and latency.

CPU-only embedding: BAAI/bge-small-en-v1.5 at 130 MB runs on CPU-only containers, keeping image size under 1.5 GB.

5.3 Limitations

- Session state and preference cache reset on container restart (Redis would fix this).
- Cold-start ranking is purely content-based with no popularity signal.
- Nigerian adaptation covers educated urban English only. Pidgin, Yoruba, Igbo, and Hausa are future work.
- Review generation and ranking reasoning remain LLM-API-dependent.

5.4 Future Work

Few-shot Pidgin English prompts using the Naija Corpus; popularity-weighted re-ranking post-processor; fine-tuning Llama-3.1-8B on Nigerian review data; explicit dislike signal accumulation in multi-turn sessions.

6 Conclusion

We present AGENTICULTURE, a dual-agent system that addresses all five BCT evaluation dimensions. Our contributions, composite preference embeddings, persona cold-start ranking, cross-domain collective vector transfer, self-consistency voting, and Nigerian cultural prompts directly close the gaps in prior AgentSociety solutions. The system is fully reproducible: two Docker containers, one `docker-compose up --build` command, and only open-source embedding models. AGENTICULTURE demonstrates that culturally-aware AI agents built without fine-tuning or new infrastructure are a practical and competitive approach to personalised AI for African markets.

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